

Physiology of the Nervous system



Lecture (1)

Membrane potentials. Types and mechanisms

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*Illustrations from (Lippincott illustrated medical
physiology)*



Back to basics...

Four types of tissues are recognized in the human body (nervous, muscular, connective and epithelial)

Two of them (nervous and muscular) are referred to as **EXCITABLE**, because they can generate action potential (AP)

Q. Why do only nervous and muscles have the ability to generate action potentials?

A. Because they possess fast sodium channels
(voltage gated)



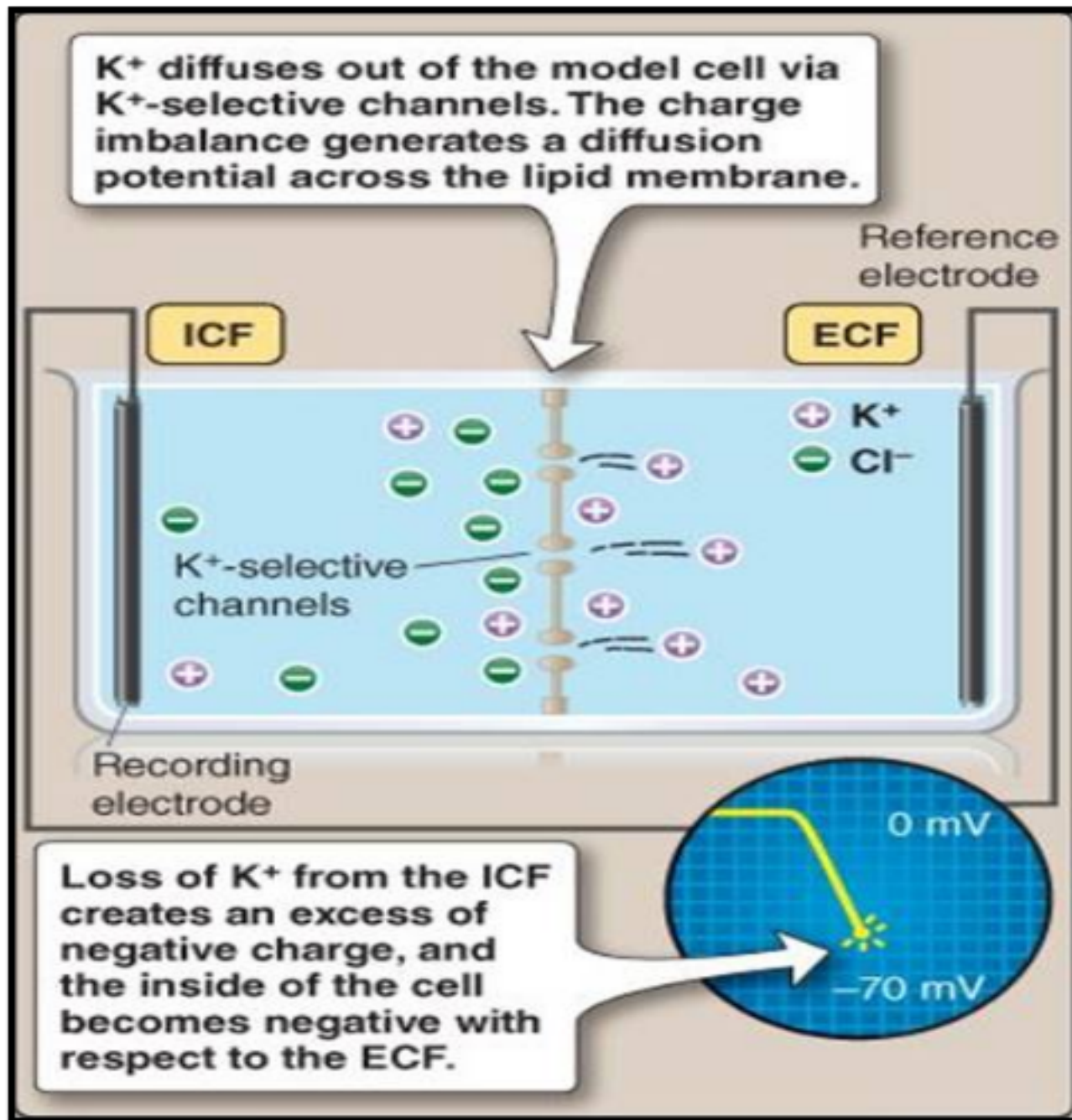
Reminder...

- Resting membrane potential (RMP) is **registered almost in all cells of body.**

- Types of membrane potentials:

- ❖ Resting membrane potential (RMP)
- ❖ Action potential (AP)
- ❖ Equilibrium potential (EP) for a specific ion



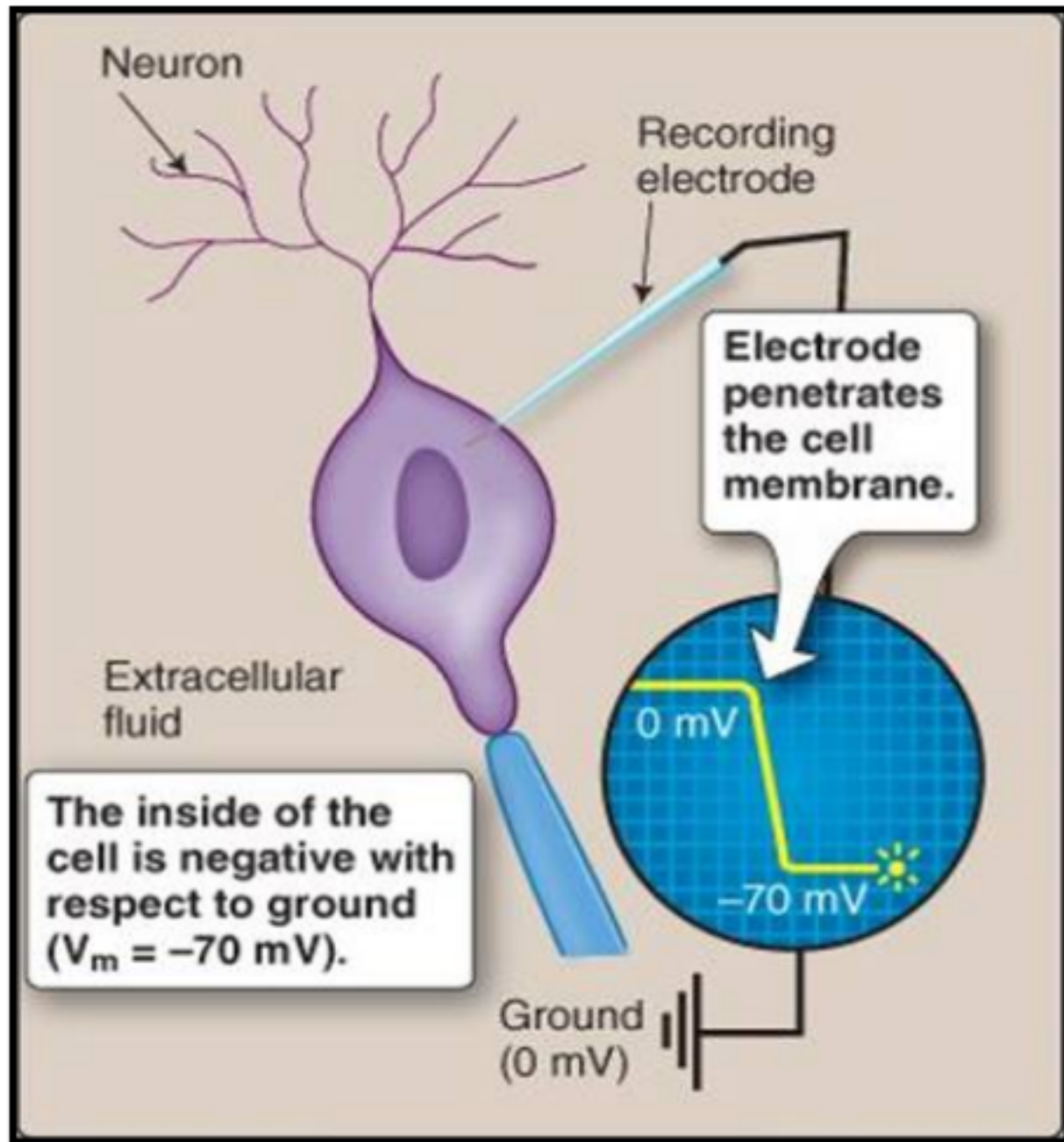


Permeability Of The
Cell Membrane (k_{efflux}
 $g(\text{potassium})$
(Conductance)

Cell membrane is
about 50-100
times more
permeable to K⁺
than it is to Na⁺

Excitable tissues (nerves and muscles)

Taking neurons as an example:
If we record the difference in voltage between the inside and the outside of the cell membrane at rest, we get RMP, which means that the negative charges inside the cell membrane are more than the negative charges outside the cell membrane



EP for the most important ions...

- Sodium **+65 mv**
- Potassium **-90 mv**
- Calcium and magnesium **+120mv**
- Chloride **-70 mv**

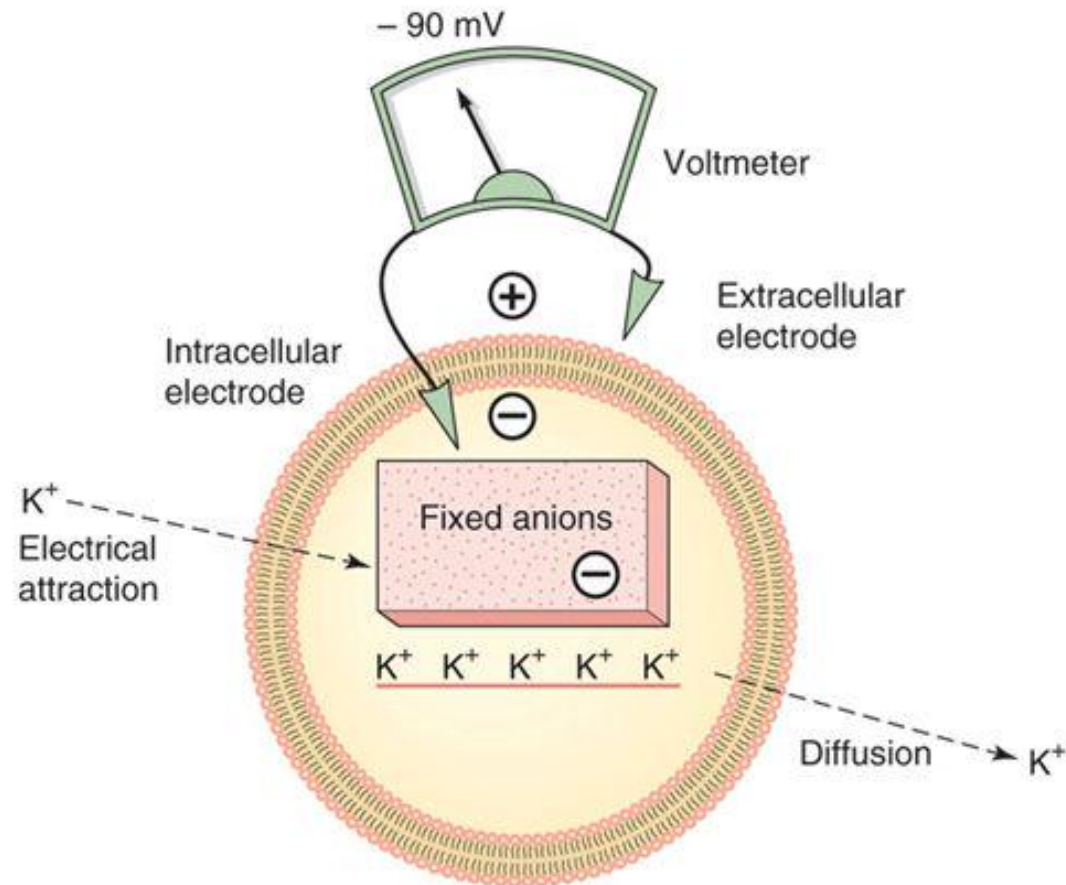
What is EP?

It is the membrane potential for an ion at which the concentration gradient and the electrochemical gradient for the ion are equal and therefore no NET movement of that ion across the cell membrane



Equilibrium Potential

- ▶ Describes voltage across cell membrane if only 1 ion could diffuse
- ▶ If membrane permeable only to K^+ , it would diffuse until it reaches its equilibrium potential (E_k)
 - ▶ K^+ is attracted inside by trapped anions but also driven out by its concentration gradient
 - ▶ At K^+ equilibrium, electrical and diffusion forces are = and opposite
 - ▶ Inside of cell has a negative charge of about -90mV



Nernst equation

$$\text{❖ } E_x = RT/F z \ln ([X]_{\text{out}}/[X]_{\text{in}})$$

❖ R = Gas constant ($8.31 \text{ J mol}^{-1} \text{ K}^{-1}$)

❖ T = Temperature (Kelvin)

❖ F = Faraday constant (96500 C mol^{-1})

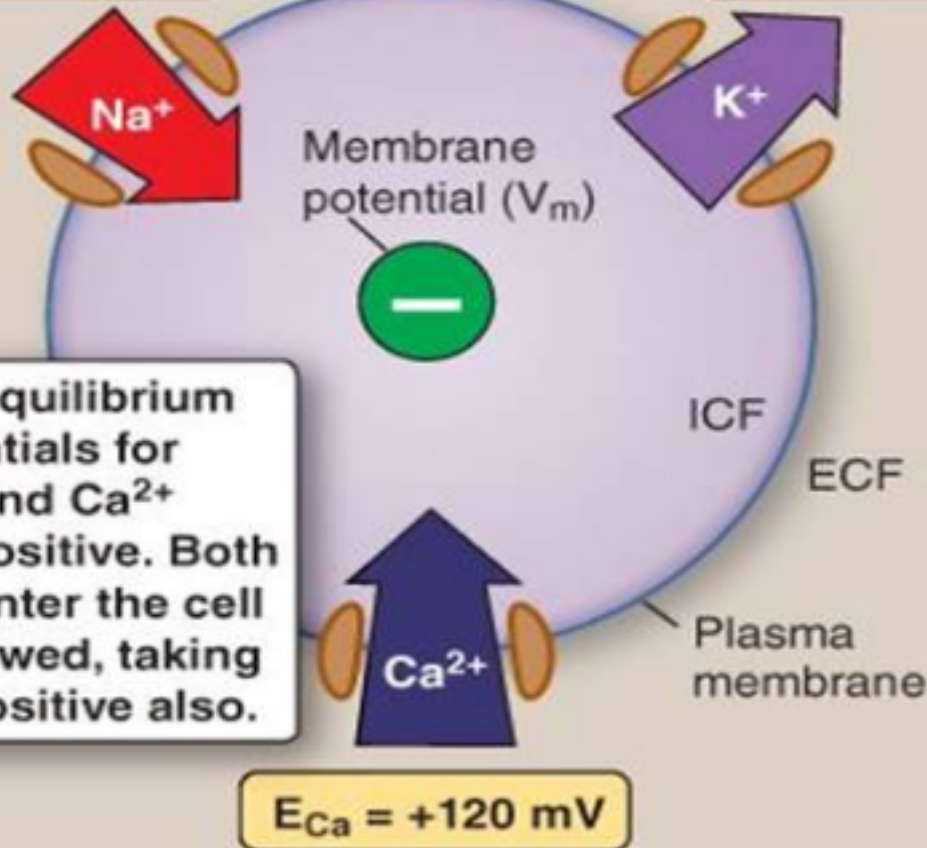
❖ Z = Ion valence



E_K is usually negative to V_m .
 K^+ will flow out of a cell until V_m reaches -90 mV, and the electrical and chemical gradients equilibrate.

$E_{Na} = +61$ mV

$E_K = -90$ mV



The equilibrium potentials for Na^+ and Ca^{2+} are positive. Both will enter the cell if allowed, taking V_m positive also.

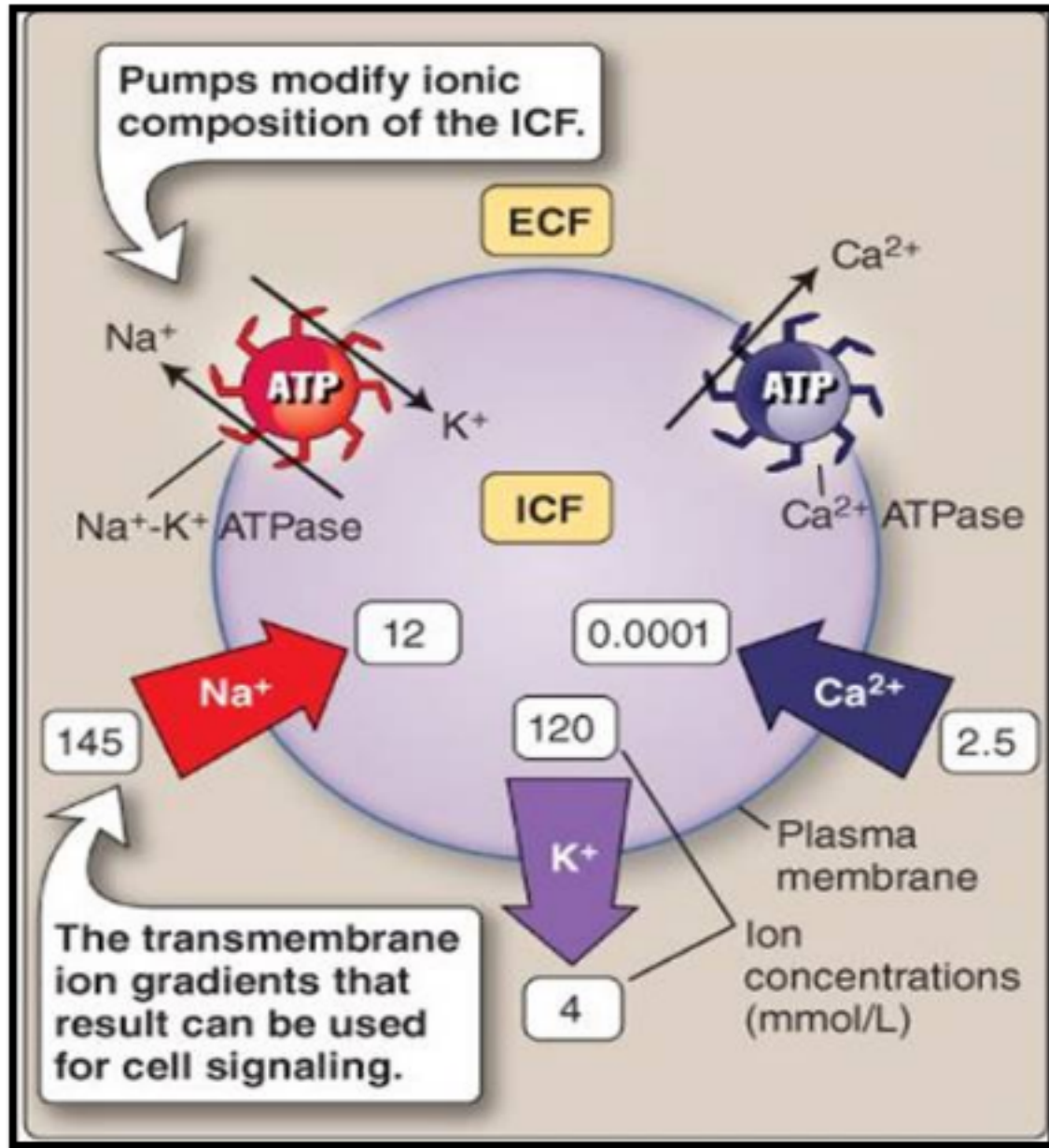
The EP for the different ions can be calculated using [Nernst's equation](#)

[Donnan's](#) effect states that there are **trapped** ions of negative charge that stay inside the cell

Resting Membrane Potential

- The difference in electrical potential across the membrane when cell is at **REST (no stimulation)**
- In neurons, the resting membrane potential is usually about -70 mV.
- Meaning, the potential inside the neuron is 70 millivolts more negative than the potential in the extracellular.





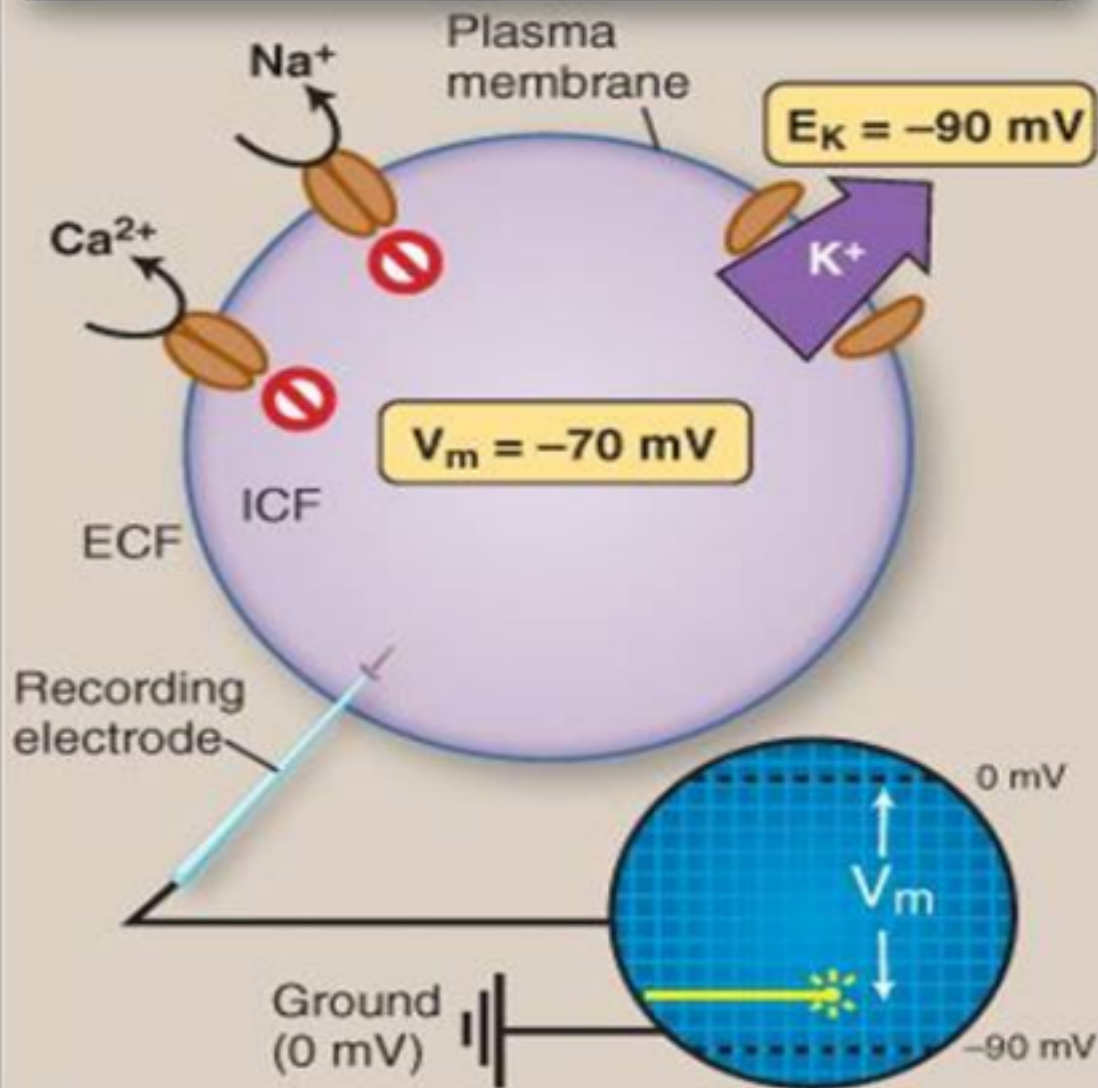
The sodium and calcium concentrations in the **ECF** is more than that in the **ICF**. The potassium concentration is the reversal.

The sodium/potassium pump keeps the concentration gradient constant by pumping out 3 sodium ions, pumping in **2** potassium ion with the consumption of **1** molecule of ATP

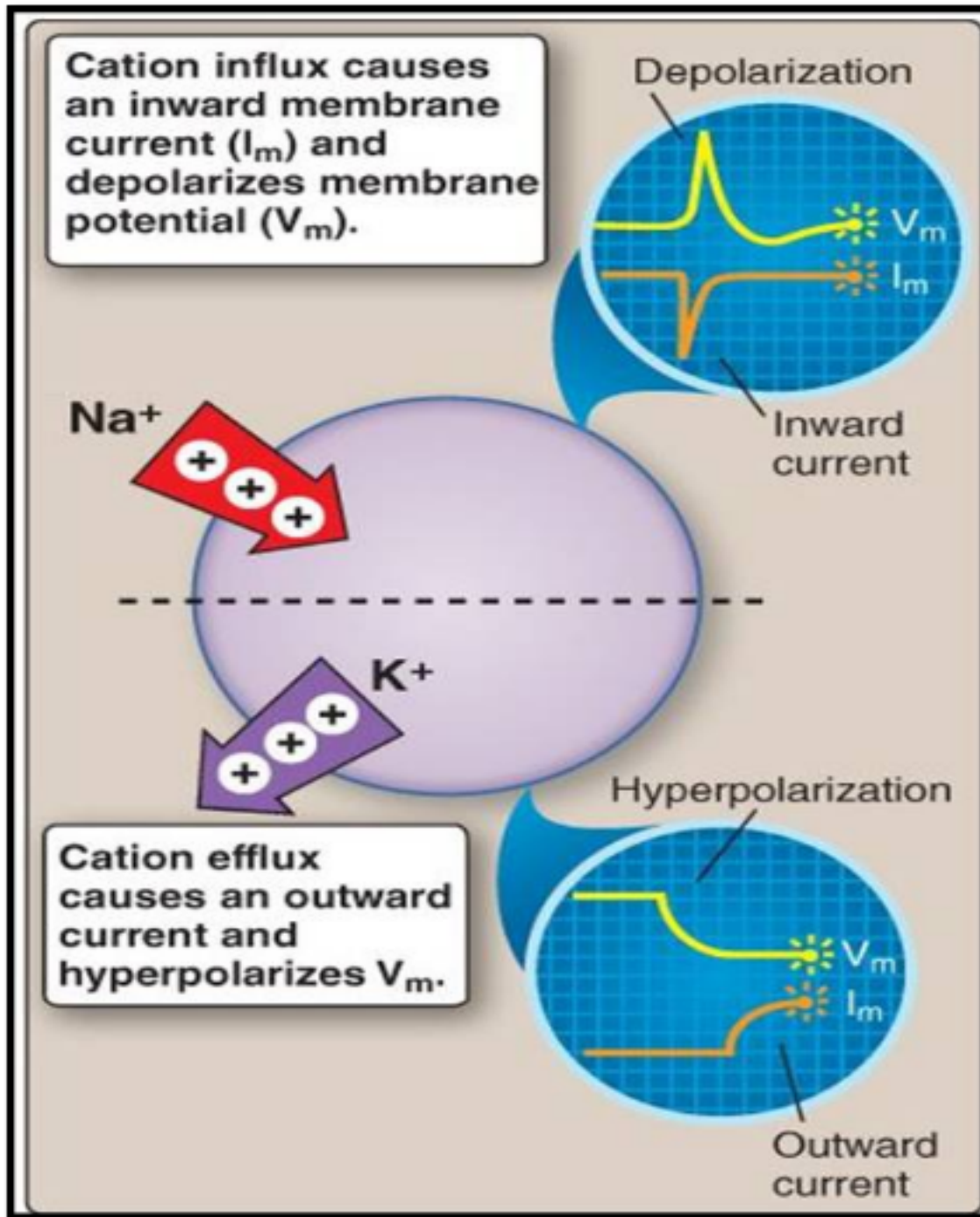
The calcium concentration gradient is kept by the

functioning calcium

Most cells have resting K^+ permeabilities, so membrane potential (V_m) lies close to the K^+ equilibrium potential (E_K).



The EP of potassium is close to the resting membrane potential because potassium has leaky channels that expel potassium out making the inside more negative. This same effect happens where the phase of hyperpolarization occurs, for the potassium ion desires to reach a state of equilibrium at **-90** mV.

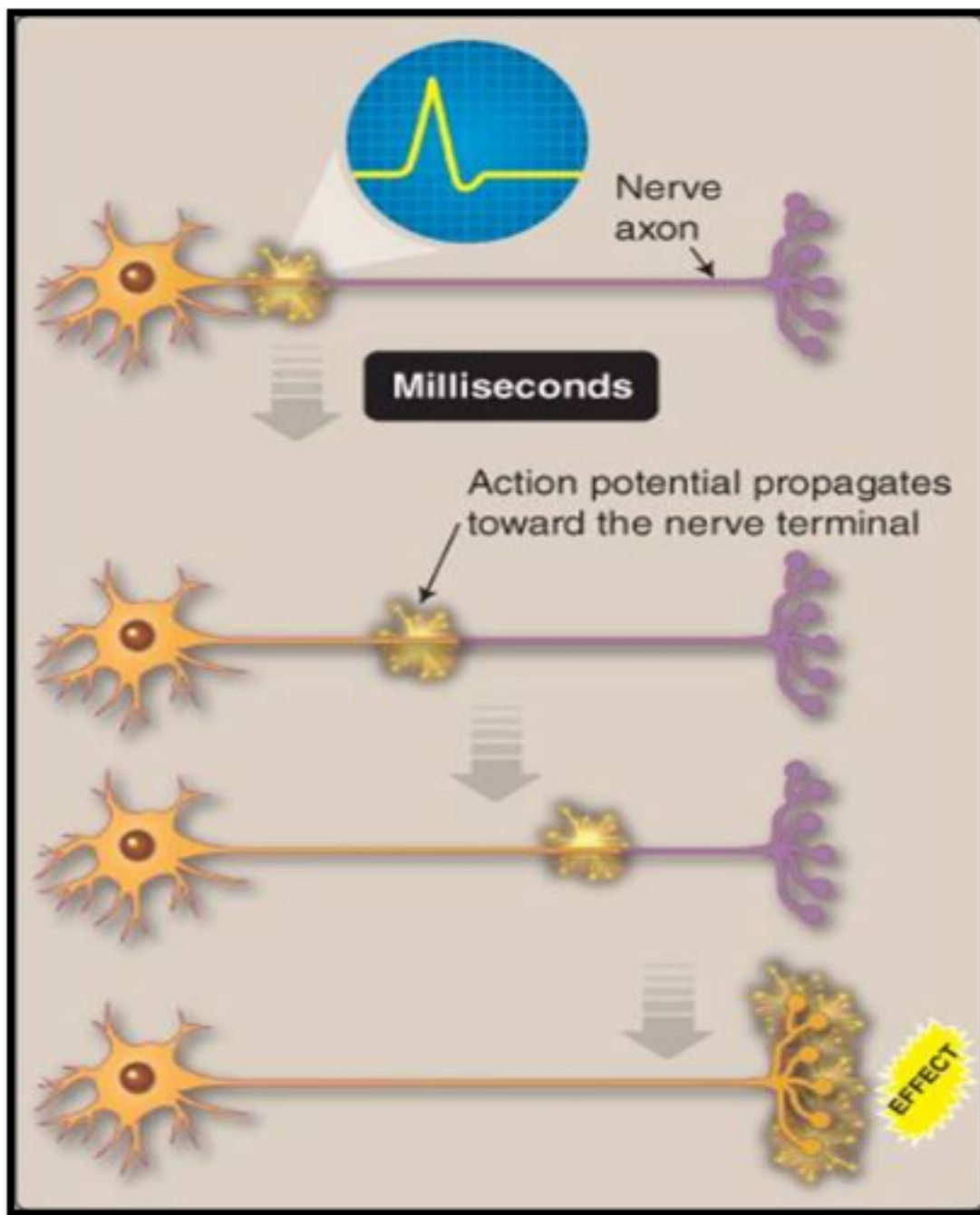


Conductance (g) of an ion through a cell membrane is the flow of the ion
 $g(\text{sodium})$
 $g(\text{potassium})$

What is an action potential?

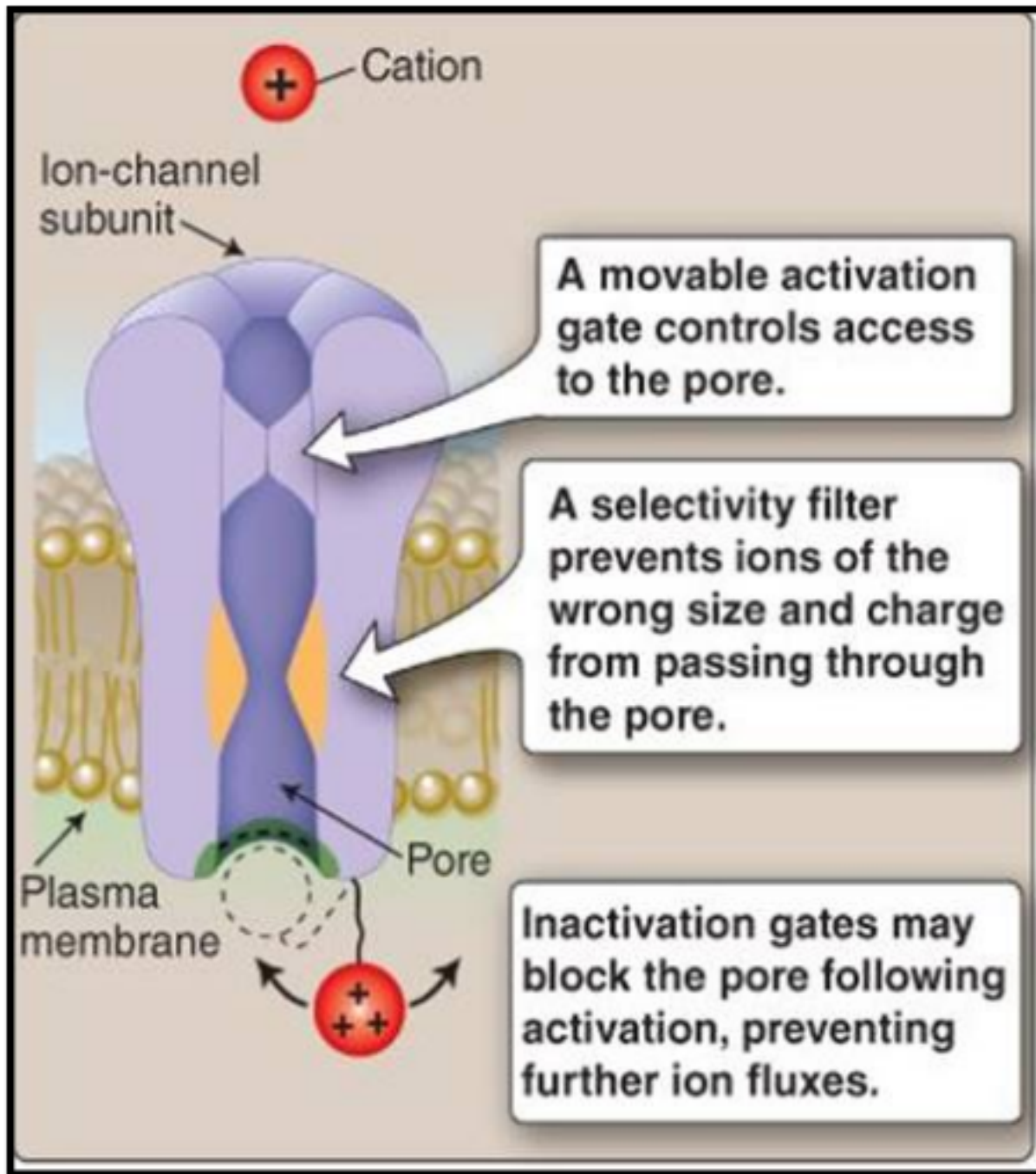
- It is the rapid, reversible change of the membrane potential due to threshold stimulation
- Mind you: the stimulation has to be a threshold, otherwise the membrane will not depolarize



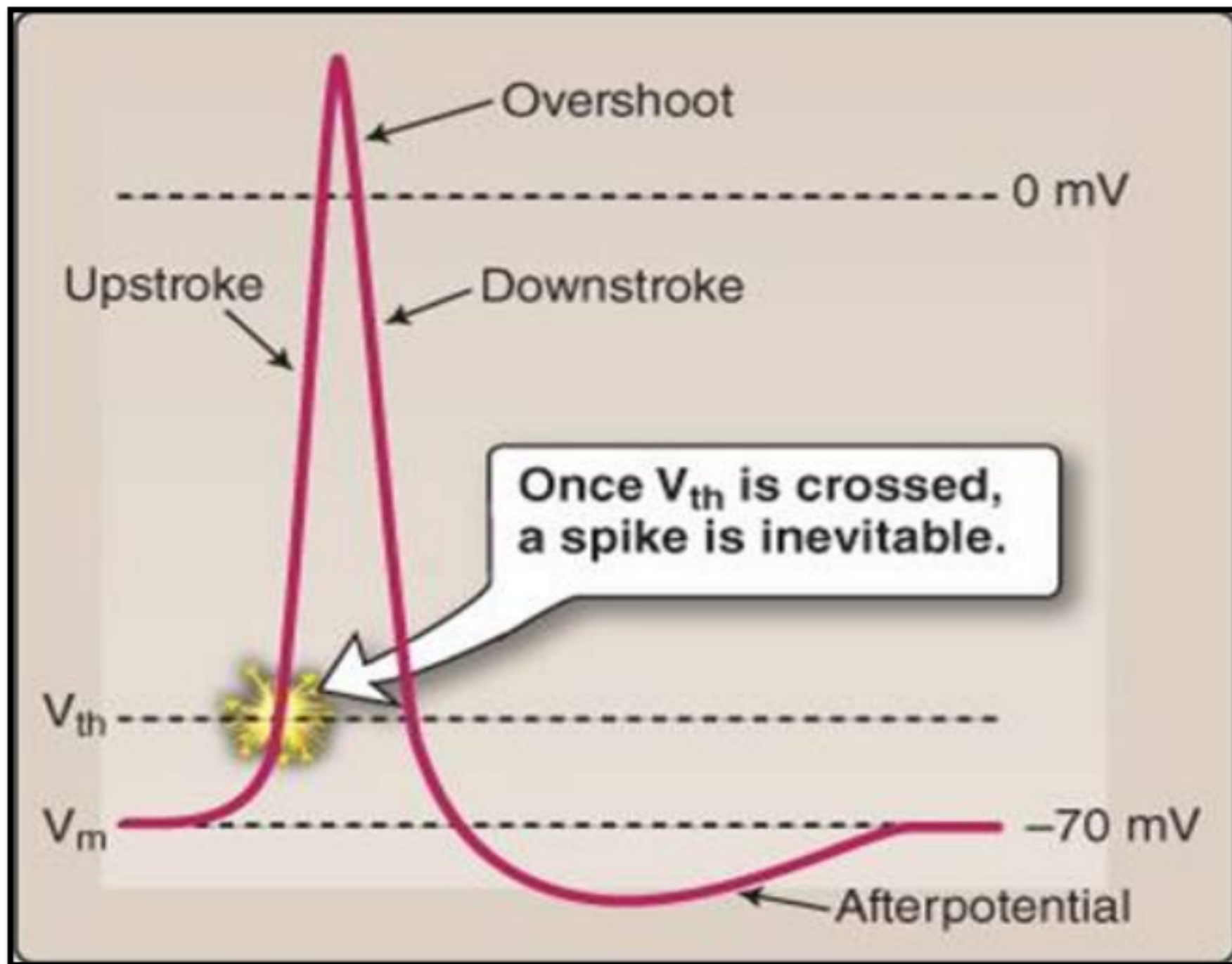


Characteristics of an action potential:

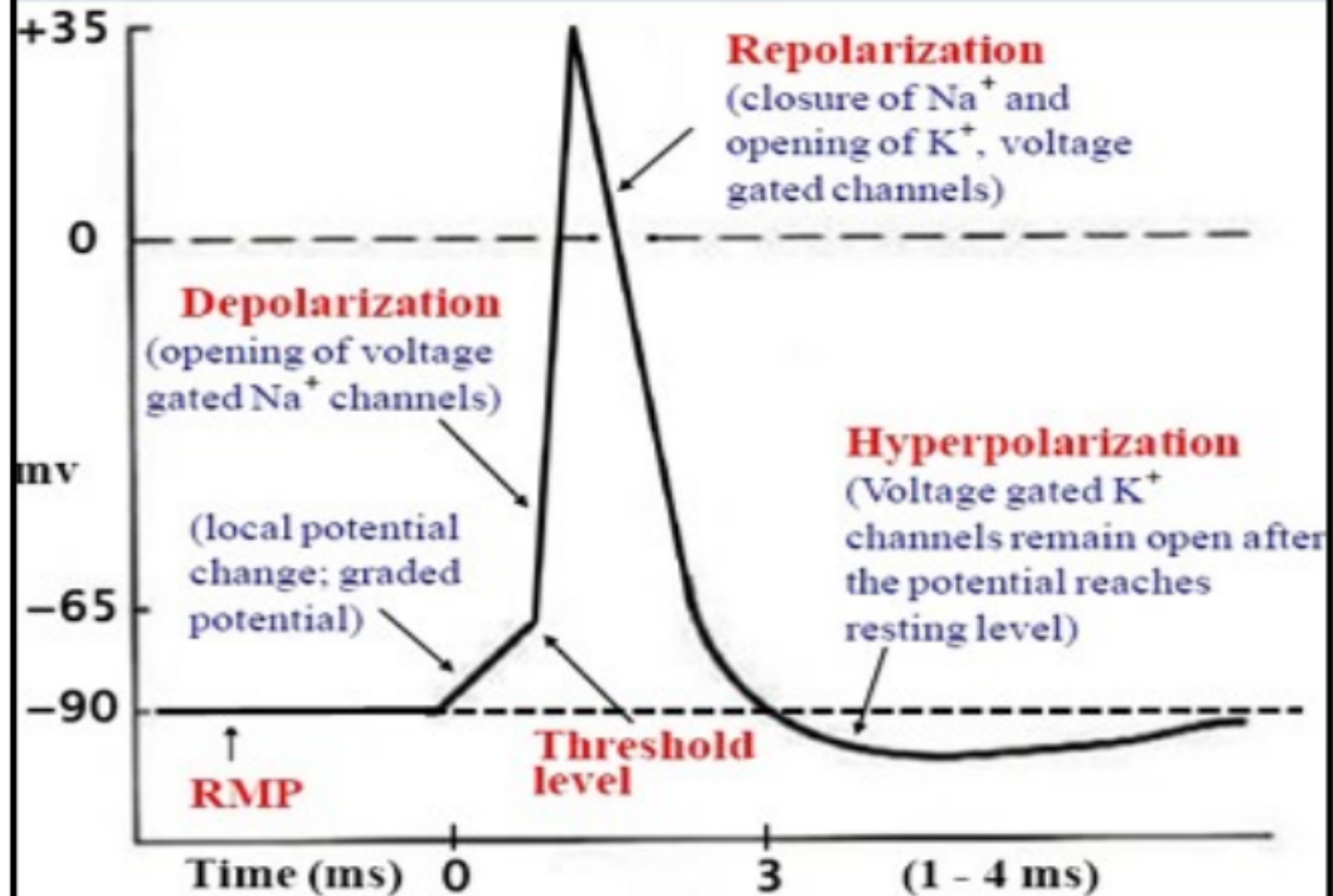
- All or none law.....
- Propagation.....
-
- Refractory period.

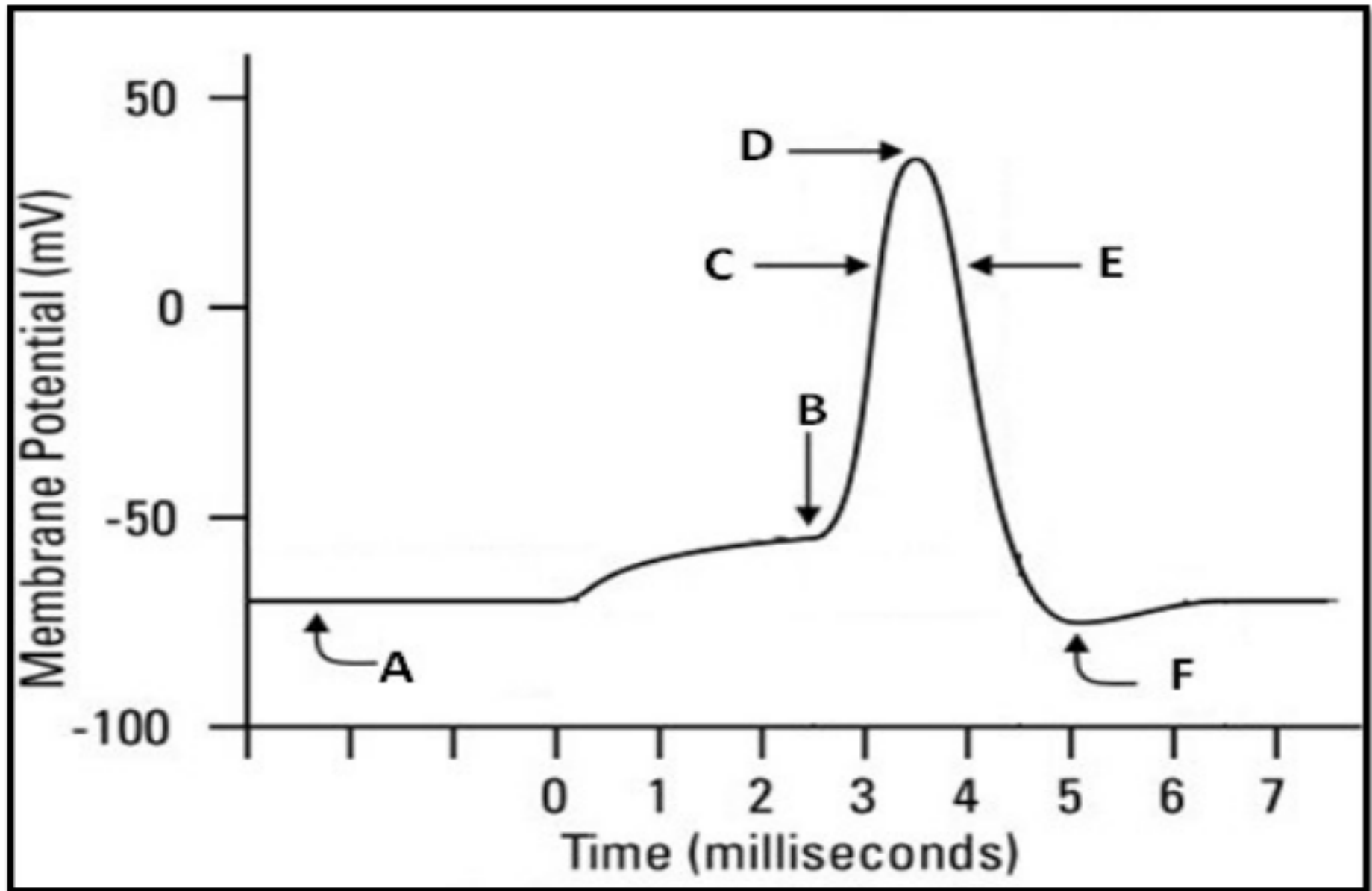


The structure of an ion channel



Action potential





Name the phases and the mechanism of genesis...



Threshold

Threshold is excitation point is also known (firing level) about -55mV

- If the force of the stimulus reaches the threshold, then the stimulus is called

threshold stimulus

opening of voltage-gated Na^+ and K^+ channels

If not reached , action potential never occur “ *All or None Law* ”



Physiology lit. (recommended)...

- Guyton and Hall textbook of physiology
- Lippincott illustrated physiology
- Ganong review of medical physiology
- Linda Costanzo textbook of medical physiology
- BRS, Linda Costanzo
- Essential to physiology

